Customer Shopping Behavior Analysis

1.Project Overview

This Project analyzes customer behavior using transactional data from 3900 purchases across various product categories the goal is to uncover the insights into spending patterns, customer segments product preferences, and behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
-columns: 18
- Key Features:
  - customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - shopping behavior (discount applied, promo code used, previous purchases, frequency of purchases, review rating, shipping type)
  - Missing data: 37 values in review rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in python:

- **Data Loading**: Imported the dataset using pandas.
- **Initial Exploration**: Used df.info() to check the structure and .describe() for summary statistics.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customer_id             3900 non-null   int64
 1   age                     3900 non-null   int64
 2   gender                  3900 non-null   object
 3   item_purchased          3900 non-null   object
 4   category                3900 non-null   object
 5   sales                   3900 non-null   int64
 6   location                3900 non-null   object
 7   size                    3900 non-null   object
 8   color                   3900 non-null   object
 9   season                  3900 non-null   object
 10  review_rating           3863 non-null   float64
 11  subscription_status     3900 non-null   object
 12  shipping_type           3900 non-null   object
 13  discount_applied        3900 non-null   object
 14  promo_code_used         3900 non-null   object
 15  previous_purchases      3900 non-null   int64
 16  payment_method          3900 non-null   object
 17  frequency_of_purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
```

	customer_id	age	gender	item_purchased	category	sales	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied	promo_code_used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

previous_purchases	payment_method	frequency_of_purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- **Missing Data Handling**: checked for null values and imputed missing values in the review rating column using the median rating of each product category.
- **Column standardization**: Renamed columns to snake casing for better readability and documentation.
- **Feature Engineering**:
  - Created age_group column by binning customer ages.
  - Created purchase_frequency_days column form purchase data.
- **Data Consistency Check**: Verified if discount_applied and promo_code_used were redundant dropped promo_code_used.
- **Database Integration**: Connected Python script to PostgreSQL and loaded the cleaned dataset into the database for SQL analysis.

4. Data Analysis using SQL

1. **Revenue by Gender** – compared total revenue generated by male vs. female customers.

	gender text	total_sales numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

3. **Top 5 Products by Rating** - Found the products with the highest average review ratings.

	item_purchased text	average_review_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping type comparison** – Compared average purchase amounts between standard and express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscriber vs. Non-subscriber** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_sales numeric	avg_sales numeric	total_customers bigint
1	No	170436.00	59.87	2847
2	Yes	62645.00	59.49	1053

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	discounted_percentage numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	count bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	category text	item_purchased text	total_orders bigint	item_rank bigint
1	Accessories	Jewelry	171	1
2	Accessories	Sunglasses	161	2
3	Accessories	Belt	161	3
4	Clothing	Blouse	171	1
5	Clothing	Pants	171	2
6	Clothing	Shirt	169	3
7	Footwear	Sandals	160	1
8	Footwear	Shoes	150	2
9	Footwear	Sneakers	145	3
10	Outerwear	Jacket	163	1
11	Outerwear	Coat	161	2

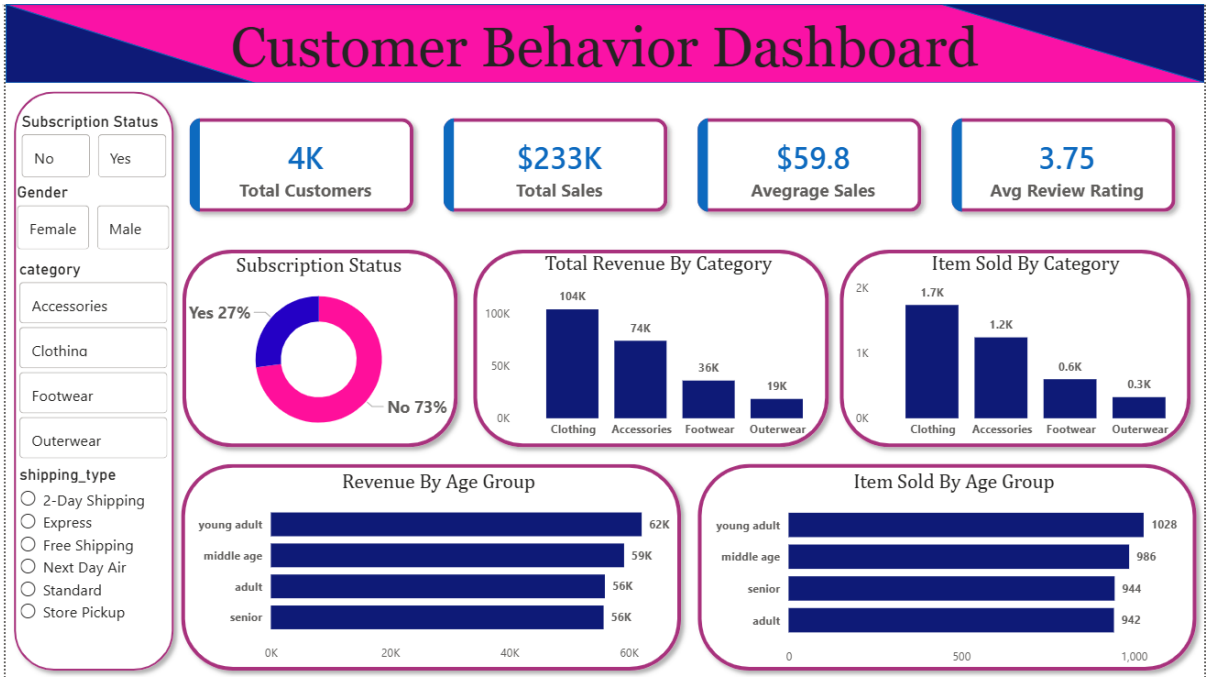
9. **Repeat buyers & Subscriptions** – Checked whether customers with < 5 purchases are more likely to subscribe.

	customers bigint	subscription_status text
1	2518	No
2	958	Yes

10. **Revenue By Age Group** – Calculated total revenue by each age group.

	age_group text	total_sales numeric
1	young adult	62143
2	middle age	59197
3	adult	55978
4	senior	55763

5. Dashboard in PowerBI



6. Business Recommendations

- **Boost Subscriptions** - Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat-buyers to move them into the "Loyal" segment/.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning –** Highlight top rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.